WPI Credential System

Oracle-Verified Blockchain Credentials for Project-Based Higher Education

---

Whitepaper v1.0
April 2026

Daniel N. Treku, Ph.D.
Assistant Teaching Professor, The Business School
Worcester Polytechnic Institute

WPI FinTech Lab

Built with Chainlink Functions | W3C Verifiable Credentials 2.0 | Open Badges 3.0

Contents

1. Introduction
2. Problem Statement
3. Solution Overview
4. Technical Architecture
5. Smart Contract Design
6. Chainlink Integration
7. Impact Assessment Framework
8. Credential Standards Compliance
9. Regulatory Context
10. Institutional Precedents
11. Roadmap
12. Team and Institutional Commitment
13. Open Source and Licensing
14. References

1. Introduction

This paper presents the design and implementation plan for an oracle-verified blockchain credential system developed at Worcester Polytechnic Institute (WPI). The system issues non-transferable (soulbound) ERC-721 tokens encoding verified project outcomes for students completing WPI's required Major Qualifying Projects (MQPs) and Interactive Qualifying Projects (IQPs). Chainlink Functions provides the oracle infrastructure for independent verification of impact claims against external data sources. The credential schema conforms to the W3C Verifiable Credentials Data Model 2.0 and 1EdTech Open Badges 3.0 specification.

WPI is the only undergraduate institution in the United States that requires every student to complete both an industry-sponsored technical capstone and a community-impact project. Together, these programs produce over 1,100 projects per year with real stakeholders and identifiable outcomes. The credential system described here instruments that existing pipeline with oracle-verified, blockchain-anchored accountability.

All code is released under the MIT license. The pilot operates at zero direct cost using open-source tools and the Ethereum Sepolia testnet.

2. Problem Statement

Higher education credentials are opaque, institution-dependent, and resistant to independent verification. A university transcript records grades. It does not record what a student built, what competencies were demonstrated, or whether the work produced measurable change for the stakeholders it was designed to serve. Employers cannot independently verify project claims. Funders cannot confirm impact. Accreditation bodies cannot aggregate outcome data across cohorts.

Existing digital credentialing platforms—Credly, Accredible, Parchment—issue assertions. They attest that an institution made a claim about a student. None verify whether the underlying claim is true. The result is a credentialing ecosystem built on institutional authority rather than evidence. This paper calls this the *verification gap*.

The verification gap is especially costly at project-based institutions. WPI students produce working prototypes, community interventions, policy analyses, and engineering deliverables. The institutional record of this work is a letter grade and a PDF report stored in an institutional repository. The gap between what students produce and what their credentials verify is the problem this system addresses.

3. Solution Overview

3.1 WPI Learning Token

The WPI Learning Token is a soulbound ERC-721 credential issued to students upon project completion. Each token stores structured metadata on-chain: project title, skills taxonomy, deliverable descriptions, partner organization, SDG alignment, and a verified impact score. The token is non-transferable by design—it remains permanently bound to the student's wallet, preventing misattribution, sale, or trade. Employers verify credentials by querying the blockchain directly through a public verification portal; no contact with WPI's registrar is

required.

3.2 Net Impact Credential

The Net Impact Credential is a graduation-level distinction—not a course—that certifies a student's cumulative demonstrated impact across their project portfolio. It aggregates impact assessment data from each qualifying project into a blockchain-verified impact transcript legible to employers, funders, policymakers, and accreditation bodies.

3.3 Oracle Verification

The distinguishing feature of this system is oracle-based impact verification. Chainlink Functions executes custom JavaScript off-chain, querying external data sources—World Bank Open Data, EPA Environmental Data Gateway, partner-organization APIs—and returning structured verification results on-chain. When a project team claims that their intervention reduced water contamination by 30 percent, the oracle cross-checks that claim against independent monitoring data. The credential records whether the claim was confirmed, not confirmed, or inconclusive. This converts the credential from an institutional assertion into an evidence-backed attestation.

4. Technical Architecture

The system consists of four layers: a smart contract layer (on-chain), an oracle layer (Chainlink), a storage layer (IPFS), and an application layer (frontend). All components are free or open-source.

Layer	Technology	Cost	Purpose
Smart contracts	Solidity 0.8.24, OpenZeppelin ERC-721	$0	Credential minting, soulbound enforcement, access control
Oracle	Chainlink Functions	$0 (testnet)	Impact claim verification against external APIs
Interoperability	Chainlink CCIP	Planned	Cross-chain credential portability
Automation	Chainlink Automation	Planned	Batch issuance, periodic re-verification
Off-chain storage	IPFS via Pinata (free tier)	$0	Full project reports, evidence documents
Frontend	Next.js, ethers.js, Vercel	$0	Admin dashboard, public verification portal
Node access	Alchemy (free tier)	$0	Sepolia RPC endpoint
Credential standard	W3C VC 2.0, Open Badges 3.0	Open standard	Interoperable credential format

Table 1. Technology stack. All components operate on free tiers during the pilot phase.

The architecture separates concerns deliberately. On-chain storage is minimal: only the credential metadata struct and verification flags. Full project reports and evidence documents are stored on IPFS, referenced by content hash. The frontend queries the blockchain directly via ethers.js—no backend server is required, eliminating hosting costs and single points of failure for the verification portal.

5. Smart Contract Design

5.1 WPICredentialRegistry

The credential registry is an ERC-721 contract with two modifications. First, soulbound enforcement: the `_update` function is overridden to revert on all transfers except minting (where `from == address(0)`). Second, structured metadata: each token ID maps to a Credential struct stored on-chain.

Credential struct:

```
struct Credential {
  string projectTitle;
  string ipfsHash;
  string skillsTags;
  string sdgAlignment;
  uint8 impactScore;   // 0–100
  uint256 issuedAt;
  bool oracleVerified;
  address verifierContract;
}
```

Access control uses a two-tier model. The contract owner (deployer) can authorize additional issuers—faculty advisors, lab administrators—who can mint credentials and update impact scores. Custom errors (`NotAuthorizedIssuer`, `SoulboundNonTransferable`, `InvalidImpactScore`) replace string-based reverts for gas efficiency.

5.2 ImpactVerifier

The ImpactVerifier contract inherits from Chainlink's `FunctionsClient`. It accepts structured impact data (metric name, claimed value, data source URL), constructs a Chainlink Functions request with inline JavaScript source, and sends it to the decentralized oracle network. The `fulfillRequest` callback decodes the oracle's response—a boolean verification flag and a uint8 score—and stores the result for the corresponding credential.

5.3 Test Coverage

The test suite covers 15 test cases across six categories: deployment, issuer management, credential issuance, soulbound enforcement, impact score updates, and view functions. Tests verify both success paths and failure modes, including unauthorized issuance attempts, transfer rejections, out-of-range score validation, and non-existent token queries.

6. Chainlink Integration

Chainlink is not a convenience dependency. It is the mechanism that closes the verification gap—the technical component that separates this system from every existing digital badge platform. Three Chainlink services are used:

6.1 Chainlink Functions

Functions provides serverless compute on Chainlink's decentralized oracle network. The ImpactVerifier contract sends a request containing custom JavaScript that queries an external API, compares the returned data against the project's claimed impact metrics, and returns a structured result on-chain. This eliminates the need for custom oracle node infrastructure and reduces the trust assumption to Chainlink's battle-tested DON. On the Sepolia testnet, Functions requests are free.

6.2 Chainlink CCIP

The Cross-Chain Interoperability Protocol enables credential portability. A credential issued on Polygon must be verifiable from Ethereum mainnet or any CCIP-supported chain. This is planned for Phase 3 (production deployment) and ensures credentials are not locked to a single network.

6.3 Chainlink Automation

Automation handles time-based and condition-based triggers: batch credential issuance at the end of each academic term, periodic impact re-verification for longitudinal tracking, and credential status finalization after advisor-review windows close. This reduces manual administrative overhead as the system scales.

7. Impact Assessment Framework

Each project team completes an Impact Assessment Report (IAR) scored across five dimensions. The aggregate score is recorded in the credential's `impactScore` field (0–100).

Dimension	Weight	Description
Stakeholder Reach	20%	Number and type of individuals, organizations, or communities directly affected
Outcome Evidence	30%	Measurable changes attributable to the project, using pre/post data where available
SDG Alignment	15%	Mapping of project outcomes to specific UN SDG targets with narrative justification
Sustainability	20%	Evidence that project outcomes will persist (handoff plan, adoption, open-source release)
Unintended Effects	15%	Honest assessment of negative externalities, displacement effects, and trade-offs

Table 2. Impact assessment dimensions. Scores of 80+ earn "Verified High Impact"; 60–79 earns "Verified Impact."

8. Credential Standards Compliance

The credential schema maps to two international standards, ensuring interoperability with existing verification infrastructure.

Standard	Version	Role in This System
W3C Verifiable Credentials	Data Model 2.0	Defines the credential's data structure, proof mechanism, and issuer/holder/verifier roles. The on-chain transaction serves as the cryptographic proof.
1EdTech Open Badges	3.0	Provides the achievement-specific schema layer. Each credential is an Open Badge whose claims are independently verifiable.
ERC-721	OpenZeppelin v5.x	Token standard for credential issuance. Modified with soulbound transfer restrictions.
ERC-5192	Minimal Soulbound	Interface for signaling non-transferability to wallets and marketplaces.

Table 3. Standards compliance.

9. Regulatory Context

9.1 GENIUS Act

The Guiding and Establishing National Innovation for U.S. Stablecoins Act was signed into law on July 18, 2025. It established the first federal regulatory framework for payment stablecoins, requiring 100 percent reserve backing with liquid assets, monthly public disclosures, and compliance with the Bank Secrecy Act. While this system does not issue stablecoins, the GENIUS Act's passage resolved the regulatory ambiguity that had discouraged U.S. institutions from experimenting with on-chain instruments. It signals federal endorsement of blockchain infrastructure as legitimate institutional tooling.

9.2 FERPA

No personally identifiable information is stored on-chain. Credentials are linked to pseudonymous Ethereum wallet addresses. Students opt in to the program, control their own wallets, and decide independently whether to share credentials with third parties. Student names, institutional IDs, and other FERPA-protected data are never recorded on the blockchain. A formal FERPA compliance review by WPI's legal counsel is planned prior to the pilot launch.

10. Institutional Precedents

Institution	Initiative	Relevance
University of Michigan	First U.S. university to accept a research grant denominated in stablecoin (Ripple RLUSD via UBRI, 2025). Faculty are building tokenized data agreements for infrastructure assets.	Demonstrates that institutional finance and compliance offices can process on-chain transactions. Michigan's wallet converts stablecoin to dollars on receipt.
MIT	Blockcerts digital diploma pilot (2017). Cryptographically signed credentials issued on the Bitcoin blockchain via Learning Machine (now Hyland Credentials).	Established technical feasibility of blockchain credentials. Students own and control their records—a principle this system adopts.
Chainlink / UNESCO	Community Grant to the UNESCO Global Education Coalition for blockchain education. Separate grant to UNICEF Innovation Fund.	Demonstrates Chainlink's institutional precedent for funding educational and social impact blockchain projects.
Malta	National issuance of blockchain-verified secondary school diplomas.	Government-scale adoption of W3C Verifiable Credentials standards.

Table 4. Institutional precedents for blockchain-based credentials.

11. Roadmap

Phase	Timeline	Key Milestones
Phase 0: Foundation	Summer 2026	Chainlink grant application submitted. Blockchain club chartered. MQP/IQP teams recruited. Impact assessment template developed.
Phase 1: Build	Fall 2026 (A/B term)	Smart contracts deployed on Sepolia. Frontend applications built. First impact assessments collected. Club workshop series launched.
Phase 2: Pilot	Spring 2027 (C/D term)	20–30 Learning Tokens issued. 10–15 Net Impact Credentials generated. Annual hackathon held. First conference paper submitted.
Phase 3: Scale	AY 2027–28	Production L2 deployment. 100+ credentials issued. Peer-reviewed publication. Accreditation body dialogue initiated.

Table 5. Four-phase rollout plan.

12. Team and Institutional Commitment

Daniel Treku, Ph.D. — Project lead and faculty advisor. Assistant Teaching Professor at WPI's Business

School. Works with the FinTech Lab and teaches the blockchain curriculum. Designed the credential system architecture and leads the Chainlink integration. Other faculty advisors would be added.

WPI FinTech Lab — Institutional home providing curriculum infrastructure, administrative support, and the connection between teaching, research, and industry partnerships. The Lab houses the blockchain course sequence that produces the student developer pipeline.

WPI Blockchain Club (launching 2026/27) — Student developer community. Weekly technical workshops, monthly industry speakers, annual hackathon co-sponsored with Chainlink. Provides sustained open-source contributions between MQP cycles. **WPI Invsetment Club** is already in operation to support this initiative.

MQP/IQP Teams AY 2026–27 — Two MQP teams (Credential Registry build; Impact Verifier engine) and two IQP teams (institutional adoption study; Global Projects Center impact assessment pilot). Recruiting from Fintech, MIS, CS, Data Science.

13. Open Source and Licensing

All code is released under the MIT license. The repository is publicly available at `github.com/dtreku/wpi-credentials` and includes production-quality Solidity contracts, a 15-case test suite, deployment and issuance scripts, and comprehensive documentation covering architecture, credential schema, impact framework, FERPA compliance, deployment procedures, and contribution guidelines.

The goal is to produce a reference implementation that any institution can fork, customize, and deploy—extending Chainlink's oracle infrastructure into the higher education credentialing vertical.

14. References

[1] W3C. Verifiable Credentials Data Model v2.0. https://www.w3.org/TR/vc-data-model-2.0/ (2024).

[2] 1EdTech. Open Badges Specification v3.0. https://www.imsglobal.org/spec/ob/v3p0 (2024).

[3] Weyl, E.G., Ohlhaver, P., Buterin, V. Decentralized Society: Finding Web3's Soul. SSRN 4105763 (2022).

[4] White House. Fact Sheet: President Donald J. Trump Signs GENIUS Act into Law. July 18, 2025.

[5] Covington & Burling LLP. The GENIUS Act Becomes Law: Key Provisions. July 25, 2025.

[6] University of Michigan, College of Engineering. Grant Paid in Cryptocurrency Is a First for U-M: A Q&A; with Peter Adriaens. August 21, 2025.

[7] Digital Credentials Consortium. Explaining Verifiable Credentials and Open Badges 3.0. October 2024.

[8] Blockcerts. Standards Alignment: W3C Verifiable Credentials and Data Integrity. https://www.blockcerts.org/guide/standard.html. (2024).

[9] Chainlink. Community Grant Program: Grants for Blockchain Developers and Researchers. https://go.chain.link/archives/community/grants.

[10] Chainlink. Chainlink in 2025: Dominance Across Onchain Finance. December 31, 2025.

[11] OpenZeppelin. Contracts v5.x Documentation. https://docs.openzeppelin.com/contracts/ (2024).

[12] United Nations. Transforming Our World: The 2030 Agenda for Sustainable Development (2015).